

Arrays Clicker Questions

#3

CQ arrays3

What does this print?

```
public static void go(int[] arr) {  
    arr[0] += 5;  
}  
  
public static void main(String[] args) {  
    int[] x = new int[1];  
    x[0] = 10;  
    go(x);  
    System.out.println(x[0]);  
}
```

A: 5

B: 10

C: 15

D: 0

E: none of the above

What does this print?

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public static void go(int[] arr) {  
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}  
  
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    int[] x = new int[1];  
    x[0] = 10;  
    go(x);  
    System.out.println(x[0]);  
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```

A: 5

B: 10

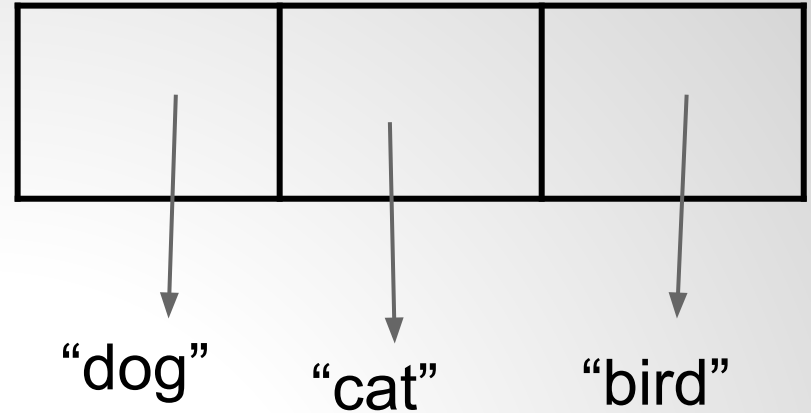
C: 15

D: 0

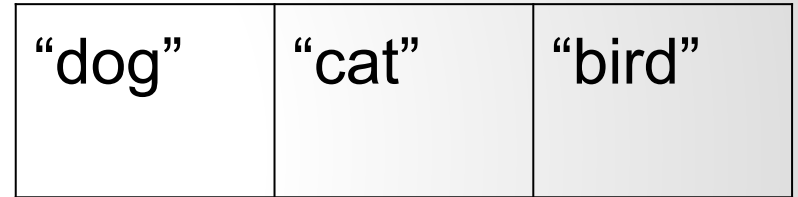
E: none of the above

Which is a more accurate depiction of an array of Strings?

A



B

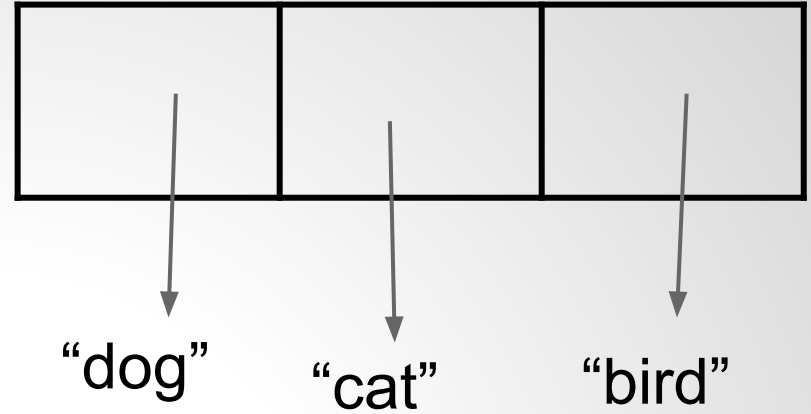


C they are exactly the same

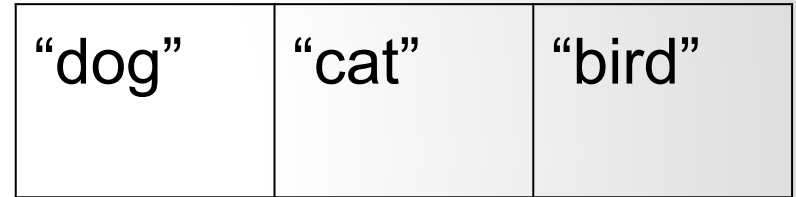
D/E none of the above
(i.e. both are wrong)

Which is a more accurate depiction of an array of Strings?

A



B

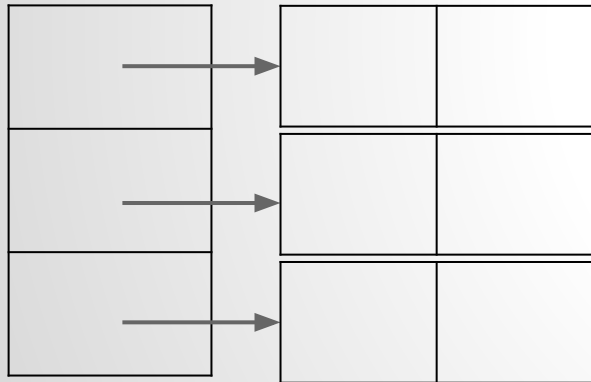


C they are exactly the same

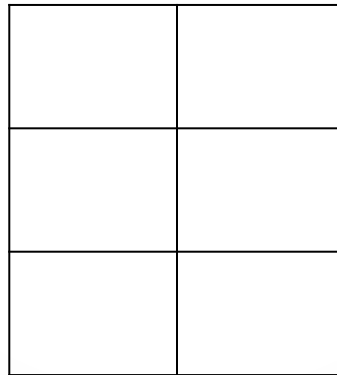
D/E none of the above
(i.e. both are wrong)

What does twoDarr look like in memory?

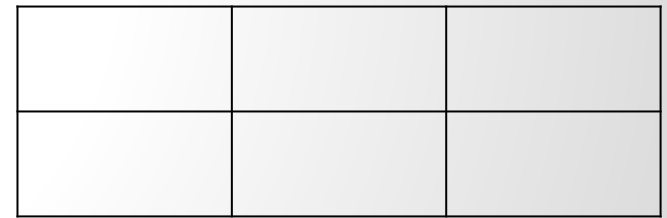
```
int[][] twoDarr = new int[3][2];
```



A



B



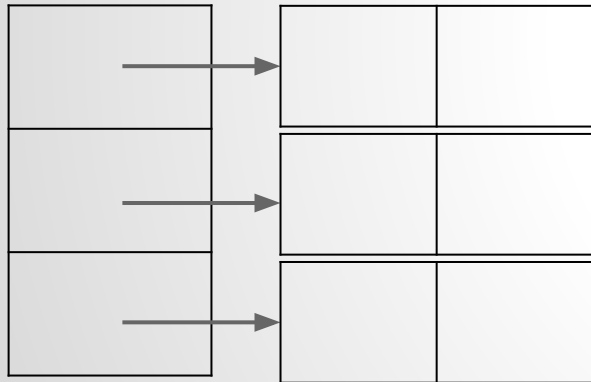
C

D all of the above

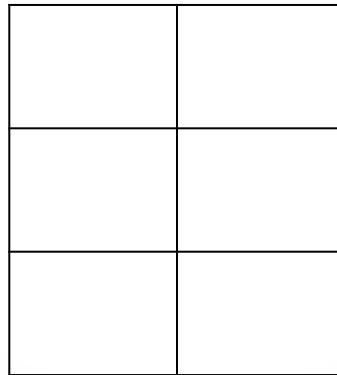
E none of the above

A is literally true, B is figuratively true

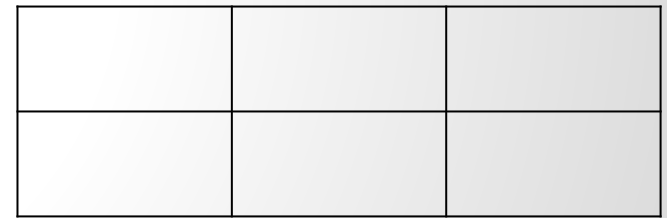
```
int[][] twoDarr = new int[3][2];
```



A



B



C

D all of the above

E none of the above

Conway's Game of Life

Very simple system of rules describe how the grid is updated between generations

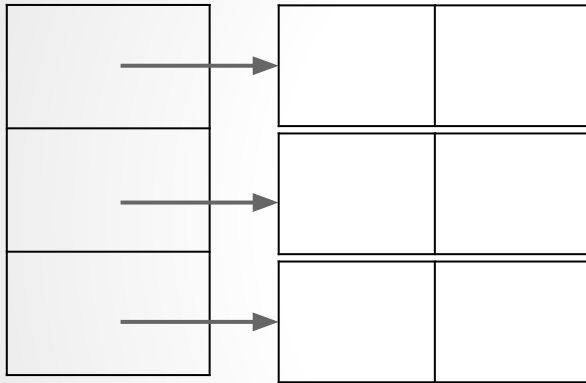
1. **Loneliness:** Any live cell with fewer than two live neighbours dies
2. **Happiness:** Any live cell with two or three live neighbours lives on to the next generation
3. **Overcrowding:** Any live cell with more than three live neighbours dies
4. **Reproduction:** Any dead cell with exactly three live neighbours becomes a live cell

Game of Life

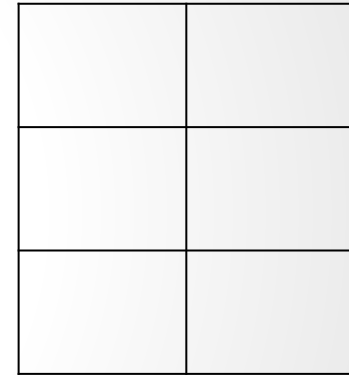
TODO: copy some slides and explain this better

TODO: show reference for twoDarr
Java implements 2D arrays as #1,
but you can think of 2D arrays as #2

```
int[][] twoDarr = new int[3][2];
```



Reality of how Java
implements 2D arrays



How we think
about 2D Java
arrays